

Estudo 2: Generalização Computacional

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1 PCA dos embeddings CLAP e MERT

```
library(tidyverse)
```

```
## Warning: pacote 'ggplot2' foi compilado no R versão 4.5.2
```

```
## Warning: pacote 'tibble' foi compilado no R versão 4.5.2
```

```
## Warning: pacote 'readr' foi compilado no R versão 4.5.2
```

```
## Warning: pacote 'purrr' foi compilado no R versão 4.5.2
```

```
## Warning: pacote 'stringr' foi compilado no R versão 4.5.2
```

```
## Warning: pacote 'forcats' foi compilado no R versão 4.5.2
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v dplyr      1.1.4      v readr      2.1.6
```

```
## v forcats    1.0.1      v stringr    1.6.0
```

```
## v ggplot2    4.0.2      v tibble     3.3.1
```

```
## v lubridate  1.9.4      v tidyr      1.3.1
```

```
## v purrr      1.2.1
```

```
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag() masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(psych)
```

```
##
## Anexando pacote: 'psych'
##
## Os seguintes objetos são mascarados por 'package:ggplot2':
##
##      %+%, alpha
```

```
library(performance)
```

```
## Warning: pacote 'performance' foi compilado no R versão 4.5.2
```

```
# 1. Carregar os dados (Ajuste os caminhos se necessário)
deam_clap <- read.csv("D:/DEAM/clap_deam_pronto_para_R.csv")
deam_mert <- read.csv("D:/DEAM/mert_deam_pronto_para_R.csv")

# 2. Preparar e Extrair Embeddings
meta_deam <- deam_mert %>% dplyr::select(file_name, valence_real, arousal_real)

# Matrizes invertidas para o PCA (Dimensões x Músicas)
mert_inv <- as.data.frame(t(deam_mert %>% dplyr::select(starts_with("X"))))
clap_inv <- as.data.frame(t(deam_clap %>% dplyr::select(starts_with("X"))))

# 3. Extrair os Componentes Principais da IA (Saliência Acústica e Multimodal)
pca_mert_deam <- principal(mert_inv, nfactors = 3, rotate = "none")
```

```
## Warning in cor.smooth(r): Matrix was not positive definite, smoothing was done
```

```
## The determinant of the smoothed correlation was zero.
## This means the objective function is not defined.
## Chi square is based upon observed residuals.
## The determinant of the smoothed correlation was zero.
## This means the objective function is not defined for the null model either.
## The Chi square is thus based upon observed correlations.
```

```
## Warning in principal(mert_inv, nfactors = 3, rotate = "none"): The matrix is
## not positive semi-definite, scores found from Structure loadings
```

```
pca_clap_deam <- principal(clap_inv, nfactors = 3, rotate = "none")
```

```
## Warning in cor.smooth(r): Matrix was not positive definite, smoothing was done
```

```
## The determinant of the smoothed correlation was zero.
## This means the objective function is not defined.
```

```

## Chi square is based upon observed residuals.
## The determinant of the smoothed correlation was zero.
## This means the objective function is not defined for the null model either.
## The Chi square is thus based upon observed correlations.

## Warning in principal(clap_inv, nfactors = 3, rotate = "none"): The matrix is
## not positive semi-definite, scores found from Structure loadings

loadings_mert <- unclass(pca_mert_deam$loadings)
loadings_clap <- unclass(pca_clap_deam$loadings)

# 4. Criar o Banco de Dados Consolidado do DEAM
df_duelo_deam <- meta_deam %>%
  mutate(
    # Embeddings MERT
    PC1_MERT = as.numeric(loadings_mert[,1]),
    PC2_MERT = as.numeric(loadings_mert[,2]),
    PC3_MERT = as.numeric(loadings_mert[,3]),

    # Embeddings CLAP
    PC1_CLAP = as.numeric(loadings_clap[,1]),
    PC2_CLAP = as.numeric(loadings_clap[,2]),
    PC3_CLAP = as.numeric(loadings_clap[,3]),

    # Criar Categorias dos Quadrantes Afetivos no DEAM (Centro = 5 na escala de 1-9)
    Category = case_when(
      valence_real >= 5 & arousal_real >= 5 ~ "Happiness",
      valence_real < 5 & arousal_real >= 5 ~ "Fear/Anger",
      valence_real < 5 & arousal_real < 5 ~ "Sadness",
      valence_real >= 5 & arousal_real < 5 ~ "Serenity"
    ),
    Category = as.factor(Category)
  )

# Ajuste de unipolaridade da Saliência (Magnitude deve ser positiva)
if(mean(df_duelo_deam$PC1_MERT) < 0) df_duelo_deam$PC1_MERT <- df_duelo_deam$PC1_MERT * -1
if(mean(df_duelo_deam$PC1_CLAP) < 0) df_duelo_deam$PC1_CLAP <- df_duelo_deam$PC1_CLAP * -1

# Definir categoria de referência no modelo para ver o contraste (Serenity será o intercepto)
df_duelo_deam$Category <- relevel(df_duelo_deam$Category, ref = "Serenity")

```

2 O DUELO DE ARQUITETURAS (2D vs HMA)

```

# --- MERT (Acústico) ---
mod_russell_mert <- lm(valence_real ~ PC2_MERT + PC3_MERT, data = df_duelo_deam)
mod_hma_mert <- lm(valence_real ~ PC1_MERT + PC2_MERT + PC3_MERT, data = df_duelo_deam)

cat("\n>>> DUELO DE PARSIMÔNIA: MERT (Acústico) <<<\n")

##
## >>> DUELO DE PARSIMÔNIA: MERT (Acústico) <<<

```

```
print(compare_performance(mod_russell_mert, mod_hma_mert, metrics = c("AIC", "R2", "RMSE")))
```

```
## # Comparison of Model Performance Indices
##
## Name          | Model | AIC (weights) | R2 | RMSE
## -----
## mod_russell_mert | lm | 5254.3 (<.001) | 0.219 | 1.037
## mod_hma_mert    | lm | 5168.3 (>.999) | 0.256 | 1.012
```

```
# --- CLAP (Multimodal) ---
mod_russell_clap <- lm(valence_real ~ PC2_CLAP + PC3_CLAP, data = df_duelo_deam)
mod_hma_clap     <- lm(valence_real ~ PC1_CLAP + PC2_CLAP + PC3_CLAP, data = df_duelo_deam)

cat("\n>>> DUELO DE PARSIMÔNIA: CLAP (Multimodal) <<<\n")
```

```
##
## >>> DUELO DE PARSIMÔNIA: CLAP (Multimodal) <<<
```

```
print(compare_performance(mod_russell_clap, mod_hma_clap, metrics = c("AIC", "R2", "RMSE")))
```

```
## # Comparison of Model Performance Indices
##
## Name          | Model | AIC (weights) | R2 | RMSE
## -----
## mod_russell_clap | lm | 5237.5 (<.001) | 0.226 | 1.033
## mod_hma_clap    | lm | 5205.4 (>.999) | 0.241 | 1.023
```

3 PROVANDO O “PARADOXO DA SERENIDADE” EM LARGA ESCALA

O modelo abaixo testa se o efeito do PC1 (Magnitude) sobre o Arousal Humano inverte ou muda drasticamente dependendo do quadrante (Interação).

```
modelo_interacao_mert <- lm(arousal_real ~ PC1_MERT * Category, data = df_duelo_deam)

cat("\n>>> TESTE DO PARADOXO DA SERENIDADE NO DEAM (MERT) <<<\n")
```

```
##
## >>> TESTE DO PARADOXO DA SERENIDADE NO DEAM (MERT) <<<
```

```
summary(modelo_interacao_mert)
```

```
##
## Call:
## lm(formula = arousal_real ~ PC1_MERT * Category, data = df_duelo_deam)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
```

```
## -1.98091 -0.49999 -0.01166 0.51375 2.11250
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      4.3435     0.9802   4.431 9.94e-06 ***
## PC1_MERT         -0.2876     1.0825  -0.266 0.79050
## CategoryFear/Anger -0.8580     1.4161  -0.606 0.54465
## CategoryHappiness   1.6315     1.1076   1.473 0.14090
## CategorySadness    -2.8613     1.1034  -2.593 0.00959 **
## PC1_MERT:CategoryFear/Anger 2.7471     1.5626   1.758 0.07892 .
## PC1_MERT:CategoryHappiness  0.3010     1.2222   0.246 0.80553
## PC1_MERT:CategorySadness  2.7243     1.2212   2.231 0.02582 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.683 on 1794 degrees of freedom
## Multiple R-squared:  0.7173, Adjusted R-squared:  0.7162
## F-statistic: 650.4 on 7 and 1794 DF,  p-value: < 2.2e-16
```

```
modelo_interacao_clap <- lm(arousal_real ~ PC1_CLAP * Category, data = df_duelo_deam)
cat("\n>>> TESTE DO PARADOXO DA SERENIDADE NO DEAM (CLAP) <<<\n")
```

```
##
## >>> TESTE DO PARADOXO DA SERENIDADE NO DEAM (CLAP) <<<
```

```
summary(modelo_interacao_clap)
```

```
##
## Call:
## lm(formula = arousal_real ~ PC1_CLAP * Category, data = df_duelo_deam)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.8495 -0.5266 -0.0270  0.5188  2.1103
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      3.2325     0.2352  13.746 < 2e-16 ***
## PC1_CLAP         1.4024     0.3808   3.682 0.000238 ***
## CategoryFear/Anger 2.5371     0.3622   7.004 3.51e-12 ***
## CategoryHappiness  2.9675     0.2996   9.905 < 2e-16 ***
## CategorySadness   -0.2948     0.2624  -1.124 0.261364
## PC1_CLAP:CategoryFear/Anger -1.4941     0.5875  -2.543 0.011069 *
## PC1_CLAP:CategoryHappiness -1.7462     0.4828  -3.616 0.000307 ***
## PC1_CLAP:CategorySadness  -0.1179     0.4306  -0.274 0.784344
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.677 on 1794 degrees of freedom
## Multiple R-squared:  0.7223, Adjusted R-squared:  0.7212
## F-statistic: 666.6 on 7 and 1794 DF,  p-value: < 2.2e-16
```

4 O EXPERIMENTO DE FALSIFICAÇÃO

4.1 IPSATIZANDO A IA (MERT) ANTES DO PCA - MERT

```
library(tidyverse)
library(psych)
library(performance)

# 1. Pegar os embeddings puros do MERT (1802 músicas x 768 dimensões)
emb_mert_puro <- deam_mert %>% dplyr::select(starts_with("X"))

# 2. A IPSATIZAÇÃO DA MÁQUINA (Row-wise scaling)
# Para cada música (linha), centralizamos as 768 dimensões na média 0 e DP 1.
# Isso destrói a magnitude acústica global do vetor.
emb_mert_ipsatizado <- t(apply(emb_mert_puro, 1, function(x) scale(x, center = TRUE, scale = TRUE)))

# 3. Transpor (como você fez no original) para o PCA extrair o peso das músicas
mert_inv_ipsat <- as.data.frame(t(emb_mert_ipsatizado))

# 4. Rodar o PCA na IA "Lobotomizada"
pca_mert_ipsatizado <- principal(mert_inv_ipsat, nfactors = 3, rotate = "none")

## Warning in cor.smooth(r): Matrix was not positive definite, smoothing was done

## The determinant of the smoothed correlation was zero.
## This means the objective function is not defined.
## Chi square is based upon observed residuals.

## The determinant of the smoothed correlation was zero.
## This means the objective function is not defined for the null model either.
## The Chi square is thus based upon observed correlations.

## Warning in principal(mert_inv_ipsat, nfactors = 3, rotate = "none"): The matrix
## is not positive semi-definite, scores found from Structure loadings

# Vamos ver o que aconteceu com a variância do PC1 da máquina?
cat("\n>>> Variância do PC1 na IA PURA vs IA IPSATIZADA <<<\n")

##
## >>> Variância do PC1 na IA PURA vs IA IPSATIZADA <<<

cat("Pura (Magnitude mantida):", pca_mert_deam$Vaccounted[2,1], "\n")

## Pura (Magnitude mantida): 0.8158053

cat("Ipsatizada (Magnitude apagada):", pca_mert_ipsatizado$Vaccounted[2,1], "\n")

## Ipsatizada (Magnitude apagada): 0.8158053
```

```

# 5. Extrair os novos loadings (sem a magnitude bruta)
loadings_ipsat <- unclass(pca_mert_ipsatizado$loadings)

df_teste_falsificacao <- meta_deam %>%
  mutate(
    # Embeddings Ipsatizados
    PC1_Ipsat = as.numeric(loadings_ipsat[,1]),
    PC2_Ipsat = as.numeric(loadings_ipsat[,2]),
    PC3_Ipsat = as.numeric(loadings_ipsat[,3]),

    # Resgatando os embeddings puros (para a batalha)
    PC1_Puro = as.numeric(loadings_mert[,1]),
    PC2_Puro = as.numeric(loadings_mert[,2]),
    PC3_Puro = as.numeric(loadings_mert[,3])
  )

# 6. O DUELO DE PREDIÇÃO (O que acontece com o Arousal?)
# Modelo A: A IA tentando prever a Ativação com os dados PUROS
mod_arousal_puro <- lm(valence_real ~ PC1_Puro + PC2_Puro + PC3_Puro, data = df_teste_falsificacao)

# Modelo B: A IA tentando prever a Ativação com os dados IPSATIZADOS
mod_arousal_ipsat <- lm(valence_real ~ PC1_Ipsat + PC2_Ipsat, data = df_teste_falsificacao)

# 7. O VEREDITO
cat("\n>>> DUELO: IA PURA vs IA IPSATIZADA (Prevendo Arousal Humano) <<<\n")

##
## >>> DUELO: IA PURA vs IA IPSATIZADA (Prevendo Arousal Humano) <<<

veredito_falsificacao <- compare_performance(mod_arousal_puro, mod_arousal_ipsat, metrics = c("AIC", "R", "RMSE"))
print(veredito_falsificacao)

## # Comparison of Model Performance Indices
##
## Name | Model | AIC (weights) | R2 | RMSE
## -----
## mod_arousal_puro | lm | 5168.3 (>.999) | 0.256 | 1.012
## mod_arousal_ipsat | lm | 5197.2 (<.001) | 0.243 | 1.021

```

4.2 IPSATIZANDO A IA (MERT) ANTES DO PCA - CLAP

```

library(tidyverse)
library(psych)
library(performance)

# 1. Pegar os embeddings puros do CLAP (1802 músicas x 512 dimensões)
emb_clap_puro <- deam_clap %>% dplyr::select(starts_with("X"))

# 2. A IPSATIZAÇÃO DA MÁQUINA (Row-wise scaling)
# Para cada música, centralizamos as 512 dimensões na média 0 e DP 1.

```

```

# Isso cega o CLAP para a magnitude/volume global da música.
emb_clap_ipsatizado <- t(apply(emb_clap_puro, 1, function(x) scale(x, center = TRUE, scale = TRUE)))

# 3. Transpor para o PCA extrair o peso das músicas
clap_inv_ipsat <- as.data.frame(t(emb_clap_ipsatizado))

# 4. Rodar o PCA na IA "Lobotomizada"
pca_clap_ipsatizado <- principal(clap_inv_ipsat, nfactors = 3, rotate = "none")

## Warning in cor.smooth(r): Matrix was not positive definite, smoothing was done

## The determinant of the smoothed correlation was zero.
## This means the objective function is not defined.
## Chi square is based upon observed residuals.

## The determinant of the smoothed correlation was zero.
## This means the objective function is not defined for the null model either.
## The Chi square is thus based upon observed correlations.

## Warning in principal(clap_inv_ipsat, nfactors = 3, rotate = "none"): The matrix
## is not positive semi-definite, scores found from Structure loadings

# Vamos inspecionar o colapso da variância do PC1
cat("\n>>> Variância do PC1 no CLAP PURO vs CLAP IPSATIZADO <<<\n")

##
## >>> Variância do PC1 no CLAP PURO vs CLAP IPSATIZADO <<<

cat("Pura (Magnitude mantida):", pca_clap_deam$Vaccounted[2,1], "\n")

## Pura (Magnitude mantida): 0.3670185

cat("Ipsatizada (Magnitude apagada):", pca_clap_ipsatizado$Vaccounted[2,1], "\n")

## Ipsatizada (Magnitude apagada): 0.3670185

# 5. Extrair os novos loadings (sem a magnitude)
loadings_clap_ipsat <- unclass(pca_clap_ipsatizado$loadings)

df_teste_falsificacao_clap <- meta_deam %>%
  mutate(
    # Embeddings Ipsatizados limitados à Lente 2D de Russell
    PC1_Ipsat = as.numeric(loadings_clap_ipsat[,1]),
    PC2_Ipsat = as.numeric(loadings_clap_ipsat[,2]),

    # Resgatando os embeddings puros (HMA - 3 Dimensões)
    PC1_Puro = as.numeric(loadings_clap[,1]),
    PC2_Puro = as.numeric(loadings_clap[,2]),
    PC3_Puro = as.numeric(loadings_clap[,3])
  )

```



```

)

# 6. O DUELO DE PREDIÇÃO (Prevendo Ativação Humana)
# Modelo A: CLAP mantendo a Magnitude + Nuances (As 3 dimensões puras)
mod_arousal_puro_clap <- lm(valence_real ~ PC1_Puro + PC2_Puro + PC3_Puro, data = df_teste_falsificacao)

# Modelo B: CLAP testado na lente 2D de Russell (Apenas as variações qualitativas)
mod_arousal_ipsat_clap <- lm(valence_real ~ PC1_Ipsat + PC2_Ipsat, data = df_teste_falsificacao_clap)

# 7. O VEREDITO
cat("\n>>> DUELO: CLAP PURO vs CLAP IPSATIZADO (Lente de Russell) <<<\n")

##
## >>> DUELO: CLAP PURO vs CLAP IPSATIZADO (Lente de Russell) <<<

veredito_falsificacao_clap <- compare_performance(mod_arousal_puro_clap, mod_arousal_ipsat_clap, metrics = c("AIC", "R2", "RMSE"))
print(veredito_falsificacao_clap)

## # Comparison of Model Performance Indices
##
## Name | Model | AIC (weights) | R2 | RMSE
## -----
## mod_arousal_puro_clap | lm | 5205.4 (>.999) | 0.241 | 1.023
## mod_arousal_ipsat_clap | lm | 5387.5 (<.001) | 0.159 | 1.076

sessionInfo()

## R version 4.5.0 (2025-04-11 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26200)
##
## Matrix products: default
## LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=Portuguese_Brazil.utf8 LC_CTYPE=Portuguese_Brazil.utf8
## [3] LC_MONETARY=Portuguese_Brazil.utf8 LC_NUMERIC=C
## [5] LC_TIME=Portuguese_Brazil.utf8
##
## time zone: America/Sao_Paulo
## tzcode source: internal
##
## attached base packages:
## [1] stats graphics grDevices utils datasets methods base
##
## other attached packages:
## [1] performance_0.15.2 psych_2.5.3 lubridate_1.9.4 forcats_1.0.1
## [5] stringr_1.6.0 dplyr_1.1.4 purrr_1.2.1 readr_2.1.6
## [9] tidyr_1.3.1 tibble_3.3.1 ggplot2_4.0.2 tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
## [1] gtable_0.3.6 compiler_4.5.0 tidyselect_1.2.1 parallel_4.5.0

```

## [5] scales_1.4.0	yaml_2.3.10	fastmap_1.2.0	lattice_0.22-7
## [9] R6_2.6.1	generics_0.1.4	knitr_1.50	insight_1.4.2
## [13] pillar_1.11.1	RColorBrewer_1.1-3	tzdb_0.5.0	rlang_1.1.6
## [17] stringi_1.8.7	xfun_0.52	S7_0.2.1	timechange_0.3.0
## [21] cli_3.6.5	withr_3.0.2	magrittr_2.0.4	digest_0.6.38
## [25] grid_4.5.0	rstudioapi_0.17.1	hms_1.1.4	nlme_3.1-168
## [29] lifecycle_1.0.5	vctrs_0.6.5	mnormt_2.1.1	evaluate_1.0.5
## [33] glue_1.8.0	farver_2.1.2	rmarkdown_2.30	tools_4.5.0
## [37] pkgconfig_2.0.3	htmltools_0.5.8.1		